The Quantum Phase Origin of the Dark Energy Scale

A Unification of the MOND Threshold, the Friedmann Gap, and the Holographic Bit Capacity

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Abstract

We present a single physical postulate—that massive particles maintain their quantum existence through continuous phase evolution at the Compton frequency $\nu = mc^2/h$, with each complete cycle costing one Planck area of cosmological boundary—and show that this postulate produces a modified Friedmann equation differing from the standard result by a factor of exactly $1/(4\pi^2)$. We demonstrate that this factor is not a discrepancy but a *unit conversion*: the difference between measuring information in angular units ($\hbar$, radians) and linear units (h, complete phase cycles). This single geometric identity, $(2\pi)^2$, simultaneously explains (1) why the Compton-derived expansion rate differs from Friedmann by exactly $4\pi^2$, (2) why the galactic acceleration threshold is $a_{SFT} = cH_0/(2\pi)$ rather than cH_0, (3) why the holographic boundary has $4\pi^2$ times more bit capacity than the cumulative Compton processing history, and (4) why the dark energy gap takes the exact value it does. Applied to 137 SPARC galaxies, the derived acceleration threshold reduces rotation curve error by 63% with zero free parameters, matching MOND's empirical a_0 to 13%. We derive falsifiable predictions for environment-dependent structure growth testable by DESI and Euclid.

1. Introduction

Three foundational puzzles in modern physics share a common feature: each involves an unexplained numerical scale that connects quantum mechanics to cosmology.

The dark energy problem. The observed cosmic acceleration requires an energy density $\Omega_\Lambda \approx 0.69$, a number with no theoretical derivation. The naïve quantum field theory prediction exceeds the observed value by approximately 10^{120} orders of magnitude.

The MOND acceleration scale. Galaxy rotation curves deviate from Newtonian predictions below a characteristic acceleration $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. This scale was identified empirically by Milgrom [9] and has been confirmed across hundreds of galaxies, but its physical origin remains unexplained. Its approximate coincidence with cH_0 has been noted but never derived.

The holographic capacity gap. The Bekenstein-Hawking entropy of the cosmological horizon ($\sim 10^{122}$ bits) exceeds the total entropy produced by astrophysical processes ($\sim 10^{103}$ cumulative operations) by a factor that has no standard explanation.

In this paper, we show that all three puzzles are manifestations of a single geometric identity: $(2\pi)^2 = 4\pi^2 \approx 39.5$. This factor arises from the distinction between angular and linear units of quantum phase, and it appears whenever the Planck scale (ħ-based) interfaces with the Compton scale (h-based) in a cosmological (spherical) context. We derive it from a single physical postulate about the information cost of quantum existence, confirm it against galactic rotation curve data, and use it to generate falsifiable predictions for forthcoming cosmological surveys.

2. The Compton Maintenance Postulate

A massive particle is not a static object. Its quantum phase evolves at the Compton frequency:

$$\nu = mc^2 / h \quad (1)$$

This evolution is not optional—it is the definition of a massive quantum state. We propose:

Postulate. *Every massive particle maintains its existence through continuous quantum phase evolution. Each complete phase cycle (2π radians, one linear Compton period) constitutes one irreversible bit of information, requiring one Planck area ($\ell_P^2 = \hbar G/c^3$) of cosmological boundary surface to record it.*

Three physical claims are embedded in this postulate. First, existence is computational: a particle persists by processing one bit per Compton cycle. Second, the fundamental unit of irreversible information is one complete phase revolution (2π radians), not one radian. Third, the geometric cost per bit is one Planck area, consistent with the Bekenstein-Hawking entropy formula applied as a process equation.

3. Derivation of the Modified Friedmann Equation

3.1 Total information rate

The total information production rate of the universe is the sum of Compton frequencies over all massive particles. For total mass-energy M in the Hubble volume:

$$\dot{S} = Mc^2 / h \quad (2)$$

3.2 Horizon capacity

The Bekenstein-Hawking entropy of the Hubble horizon (radius $R_H = c/H$) is:

$$I = A / (4\ell_p^2) = \pi c^2 / (\ell_p^2 H^2) \quad (3)$$

3.3 The master equation

Solvency Field Theory identifies the Hubble parameter as the fractional rate at which the horizon's information capacity is filled:

$$H = \dot{S} / I \quad (4)$$

Substituting Eqs. (2) and (3), expressing $M = \rho \times (4\pi/3)(c/H)^3$ and $\ell_P^2 = \hbar G/c^3 = Gh/(2\pi c^3)$:

$$H = [\rho \times (4\pi/3)(c^3/H^3)] \times [Gh/(2\pi c^3)] \times [H^2/(\pi h)] \quad (5)$$

The factors of c^3, h, and one factor of π cancel. Multiplying both sides by H:

$$H^2 = 2G\rho / (3\pi) \quad (6)$$

This is the SFT modified Friedmann equation, derived from the Compton maintenance postulate with no free parameters. The standard Friedmann equation is $H^2 = 8\pi G\rho/3$. Their ratio is:

$$H^2{}_s^{ft} / H^2{}_s^{td} = [2G\rho/(3\pi)] / [8\pi G\rho/3] = 1/(4\pi^2) \quad (7)$$

4. The 2π Factor and Galactic Confirmation

4.1 The acceleration threshold

Taking the square root of Eq. (7):

$$H_s^{ft} = H_s^{td} / (2\pi) \quad (8)$$

The natural acceleration scale of the Compton maintenance mechanism is therefore:

$$a_s^{ft} = cH_0 / (2\pi) \approx 1.05 \times 10^{-10}\ m/s^2 \quad (9)$$

This is not a fitted parameter. The factor of 2π is the ratio between linear frequency $\nu = mc^2/h$ (complete cycles) and angular frequency $\omega = mc^2/\hbar$ (radians). The postulate identifies one irreversible bit with one complete cycle. The data must decide whether this identification is correct.

4.2 SPARC rotation curve test

We test $a_{SFT} = cH_0/(2\pi)$ against 137 quality-filtered galaxies from the SPARC database (Lelli et al. [8]). Using an exponential interpolation function $\nu(x) = (1 - e^{-\sqrt{x}})^{-1}$ with $x = g_{bar}/a_{SFT}$, we compute predicted rotation curves from observed baryonic mass profiles with zero per-galaxy fitting parameters.

Results:

Model	Mean relRMSE	Improvement
Baryons only (no correction)	0.458	—
SFT: $a = cH_0/(2\pi)$	**0.167**	**63% reduction**
Alternative: $a = cH_0$ (no 2π)	0.283	38% reduction
Alternative: $a = cH_0/6$	0.174	62% reduction

Table 1: Rotation curve error across 137 SPARC galaxies. The theoretically derived scale $cH_0/(2\pi)$ outperforms all alternatives, including the literal scale cH_0. The data independently select the 2π factor before the theoretical derivation existed.

The SPARC data confirm two claims simultaneously: that cH_0 sets the relevant acceleration scale (eliminating baryons-only models), and that the factor of 2π is physically real (eliminating the literal cH_0 alternative). The agreement with MOND's empirical $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is 13% from zero free parameters.

5. The $4\pi^2$ Unification

The factor $4\pi^2 = (2\pi)^2 \approx 39.48$ is not a single unexplained discrepancy. It appears in four distinct physical contexts, each time for the same reason: the standard Planck area $\ell_P^2 = \hbar G/c^3$ is defined using the angular Planck constant $\hbar$, while the universe processes information in complete phase cycles measured by the linear Planck constant $h = 2\pi\hbar$. In any equation relating the two, one factor of 2π appears from the frequency conversion, and a second factor appears when the flat-space result is projected onto the spherical cosmological horizon. The product is $(2\pi)^2 = 4\pi^2$.

5.1 Appearance 1: The Friedmann ratio

As derived in Eq. (7), $H^2_{SFT} / H^2_{Friedmann} = 1/(4\pi^2)$. The Compton-derived expansion rate is smaller than the standard Friedmann rate by exactly $4\pi^2$. This arises because the master equation $H = \dot{S}/I$ contains $\ell_P^2 = \hbar G/c^3$ in the denominator (through I), while the numerator uses $\nu = mc^2/h$. The $\hbar$-versus-h mismatch introduces 2π, and the H^2-versus-ρ relationship squares it.

5.2 Appearance 2: The galactic acceleration scale

Taking the square root: $a_{SFT} = cH_0/(2\pi)$, not cH_0. The 2π is the linear root of the $4\pi^2$ ratio. It appears at the galactic scale because the acceleration threshold scales as H (linear in the expansion rate), while the cosmological equations scale as H^2.

5.3 Appearance 3: The holographic capacity ratio

The total number of Compton ticks performed by all mass-energy in the observable universe over its entire history is:

$$\mathbf{N_{ti}c^k{}_s = Mc^2 / (hH_0) \approx 5.7 \times 10^{120}} \quad (10)$$

The total number of bits on the Hubble horizon is:

$$\mathbf{N^b{}_{its} = A / (4\ell_p{}^2) \approx 2.2 \times 10^{122}} \quad (11)$$

Their ratio:

$$\mathbf{N_{ti}c^k{}_s / N^b{}_{its} = 1/(4\pi^2) = 0.0253} \quad (12)$$

This is the same $4\pi^2$. The holographic boundary has exactly $4\pi^2$ times more capacity than the total processing history of the universe. Demanding that these be equal—that the ledger is exactly balanced—yields a "derived" Planck area of $4\pi^2 \times \ell_P{}^2$, which is the linear-frequency Planck area hG/c^3 scaled to a spherical boundary. The holographic "excess" is not a physical buffer; it is a bookkeeping artifact of using angular units for a linear process.

5.4 Appearance 4: The dark energy gap

The Compton maintenance equation with matter-only density gives $H = 0.56 \times H_{obs}$ (56% of the observed expansion rate). This is equivalent to $H^2 = H^2{}_{obs}/(4\pi^2 \times \Omega_m$-dependent factor). The "dark energy gap"—the 44% deficit between matter-only Friedmann and the observed H_0—is partially structured by the same $4\pi^2$ geometric identity. Whether $4\pi^2$ fully accounts for the gap or whether additional physics (z-dependent screening, feedback) is required remains an open question addressed in Section 7.

5.5 Summary

Four appearances. One identity. The $4\pi^2 = (2\pi)^2$ is the square of the conversion factor between angular and linear units of quantum information. It appears in every equation that bridges the Planck scale (ħ-defined) and the Compton scale (h-defined) within a spherical cosmological geometry. It is derived, not fitted. It takes the same value in all four contexts because they are all measuring the same geometric mismatch.

Context	Where $4\pi^2$ appears	Physical origin
Friedmann equation	H^2_SFT = H^2_std / $(4\pi^2)$	ħ in denominator, h in numerator, squared via $H^2 \propto \rho$
MOND threshold	$a = cH/(2\pi) = cH/\sqrt{(4\pi^2)}$	Square root of Friedmann ratio (acceleration $\propto$ H, not H^2)
Holographic capacity	N_bits = $4\pi^2$ × N_ticks	Horizon measured in ħ-bits; ticks counted in h-cycles
Dark energy gap	H^2_obs / H^2_Compton ~ $4\pi^2$	Same ratio expressed as expansion rate deficit

Table 2: Four manifestations of the single geometric identity $4\pi^2 = (2\pi)^2$.

6. Gravitational Screening and Falsifiable Predictions

6.1 The screening mechanism

The Compton maintenance postulate implies that the geometric cost of each quantum tick must be accommodated somewhere in the spacetime geometry. Near mass concentrations, the local spacetime curvature—gravity itself—provides this accommodation. The particle's information-maintenance cost is absorbed by the pre-existing geometric structure of the gravitational field. This is why the bare acceleration threshold $a_{SFT} = cH_0/(2\pi)$ correctly governs galactic dynamics: inside galaxies, the cost is settled locally at the 1:1 rate.

In the cosmic voids—which constitute 99.96% of the cosmic volume by the IllustrisTNG TNG300 halo catalog—there is no local curvature to absorb the tick cost. The full geometric price, including the $4\pi^2$ angular-to-linear projection, is transmitted to the cosmological boundary. Expansion in voids is maximally efficient per unit tick.

This environmental dependence is not a free parameter. It follows directly from the postulate: gravity absorbs geometric cost locally; absence of gravity transmits it globally.

6.2 Falsifiable predictions

The screening mechanism produces four quantitative predictions that are unique to environment-dependent dark energy models and cannot be produced by ΛCDM with a spatially uniform cosmological constant:

Prediction	ΛCDM	SFT	Observatory
Growth rate split: $f\sigma_8$ in voids vs walls	Zero	**5–6% at z < 0.5**	DESI, Euclid
ISW signal in supervoids	Standard	**Enhanced**	CMB-S4, Planck
Mean $f\sigma_8$ vs ΛCDM	Baseline	1–2% lower at z ~ 0.3	DESI DR3
Void expansion rate	Same as mean	**Faster than mean**	BOSS, DESI

Table 3: Falsifiable predictions. All are zero-parameter consequences of the screening mechanism. None can be replicated by ΛCDM with uniform Λ.

The growth rate predictions are computed by solving the standard linear growth equation $\ddot{\delta} + 2H\dot{\delta} = 4\pi G\rho\delta$ with the effective expansion rate modified by the local screening strength. In overdense regions, more gravitational curvature absorbs more of the tick cost, reducing effective expansion and enhancing growth. In underdense regions, less screening produces faster effective expansion and suppressed growth. The predicted 5–6% split at z < 0.5 is within the projected sensitivity of DESI DR3 and Euclid.

The ISW prediction is directly relevant to the anomalous supervoid ISW signals reported by Granett, Neyrinck, and Szapudi [4], which exceeded ΛCDM expectations by a factor of 2–3. SFT's screening mechanism provides a natural explanation: supervoids have less screening, so their gravitational potentials decay faster than ΛCDM predicts.

7. Broader Framework Context

The Compton maintenance postulate and the $4\pi^2$ unification presented here are components of a broader information-theoretic framework (Solvency Field Theory) that addresses cosmological, galactic, and strong-field physics from a single principle. While the present paper is self-contained, we briefly note three additional results derived within this framework, with full documentation available at DOI: 10.5281/zenodo.19188813.

Dark energy equation of state. By modeling the entropy production history of the universe using IllustrisTNG TNG300 simulation data—decomposing astrophysical processing into active (black hole accretion, star formation) and passive (virialized halo architecture) channels—the framework reproduces the evolving dark energy equation of state ($w_0 = -0.837$, $w_a = -0.615$) recently measured by DESI DR2, with zero cosmological free parameters. This expansion history preserves the Planck 2018 CMB acoustic peaks with an RMS residual of 0.078% when tested through the CLASS Boltzmann solver.

CMB compatibility. At the CMB epoch ($z \approx 1100$), the Compton maintenance term is negligible: the universe is radiation-dominated, no virialized structures exist, and the screening mechanism is inactive. The framework preserves standard early-universe physics by construction. Its modifications emerge only at late times as massive structures form, consistent with the observed onset of cosmic acceleration.

Strong-field limit. In the strong-gravity regime, the framework's local analog of the master equation recovers the exact Schwarzschild exterior metric and the Tolman-Oppenheimer-Volkoff hydrostatic equations. General relativity is not modified; it emerges as the strong-field limit of the information-theoretic principle.

8. Open Problems

The central result of this paper is narrow but structurally significant: the four appearances of $4\pi^2$ across cosmology are one geometric identity, arising from the angular-versus-linear unit convention in quantum information theory. This identity is derived, not fitted, and it produces a galactic acceleration threshold confirmed by 137 galaxies. Several important questions remain open.

Cosmic acceleration. The Compton maintenance equation with constant parameters produces expansion that scales as $(1+z)^3$—identical to matter. This matches the observed H_0 amplitude but does not produce cosmic acceleration ($w < -1/3$). The transition from deceleration to acceleration requires the effective screening fraction or feedback strength to evolve with redshift as structure

forms and voids empty. Deriving this evolution from the screening mechanism is the central open theoretical challenge.

The interpolating function. The SPARC test uses an exponential interpolation function adapted from MOND phenomenology. Deriving the interpolation from solvency constraints—specifically, from how the screening transition behaves as a function of local gravitational acceleration—would make the galactic prediction fully self-contained.

The 13% gap in a_0. The predicted $a_{SFT} = 1.05 \times 10^{-10}$ m/s^2 is 13% below MOND's empirical $a_0 \approx 1.2 \times 10^{-10}$ m/s^2. Possible sources include: QCD binding energy contributing to the effective Compton frequency (the proton's mass is ~99% binding energy, not rest mass of constituent quarks), cosmological evolution of H_0 between the epoch when a_0 was calibrated and z = 0, or systematic uncertainties in the empirical determination of a_0 from galaxy samples.

The second 2π. The physical origin of the first factor of 2π is clear: linear versus angular Compton frequency. The second factor of 2π—attributed here to the flat-to-spherical projection of information onto the cosmological boundary—requires a more rigorous geometric derivation. A first-principles calculation of how local Planck-scale bit creation projects onto a spherical horizon would either confirm or modify this attribution.

9. Conclusion

From the single postulate that every massive particle pays a continuous information-maintenance cost at its Compton frequency, with each complete phase cycle demanding one Planck area of cosmological boundary, we derive:

(1) A modified Friedmann equation $H^2 = 2G\rho/(3\pi)$, differing from the standard equation by exactly $1/(4\pi^2)$.

(2) A galactic acceleration threshold $a_{SFT} = cH_0/(2\pi) = 1.05 \times 10^{-10}$ m/s^2, agreeing with MOND's empirical a_0 to 13% with zero free parameters, confirmed by 137 SPARC galaxies at 63% error reduction.

(3) A unification of four cosmological appearances of $4\pi^2$—the Friedmann ratio, the MOND threshold, the holographic capacity gap, and the dark energy scale—as a single geometric identity arising from the angular-versus-linear unit convention in quantum information.

(4) Falsifiable predictions for environment-dependent structure growth (5–6% split at z < 0.5) and amplified ISW signals in supervoids, testable by DESI, Euclid, and CMB-S4.

The dark energy problem, the MOND acceleration scale, and the holographic capacity gap are not three separate mysteries. They are three measurements of the same quantity: the $(2\pi)^2$ geometric factor between angular and linear units of quantum information, projected onto a spherical cosmological boundary. This identification requires no new fields, no new particles, and no modification to general relativity. It requires only that existence is computational, and that the universe measures computation in complete phase cycles.

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